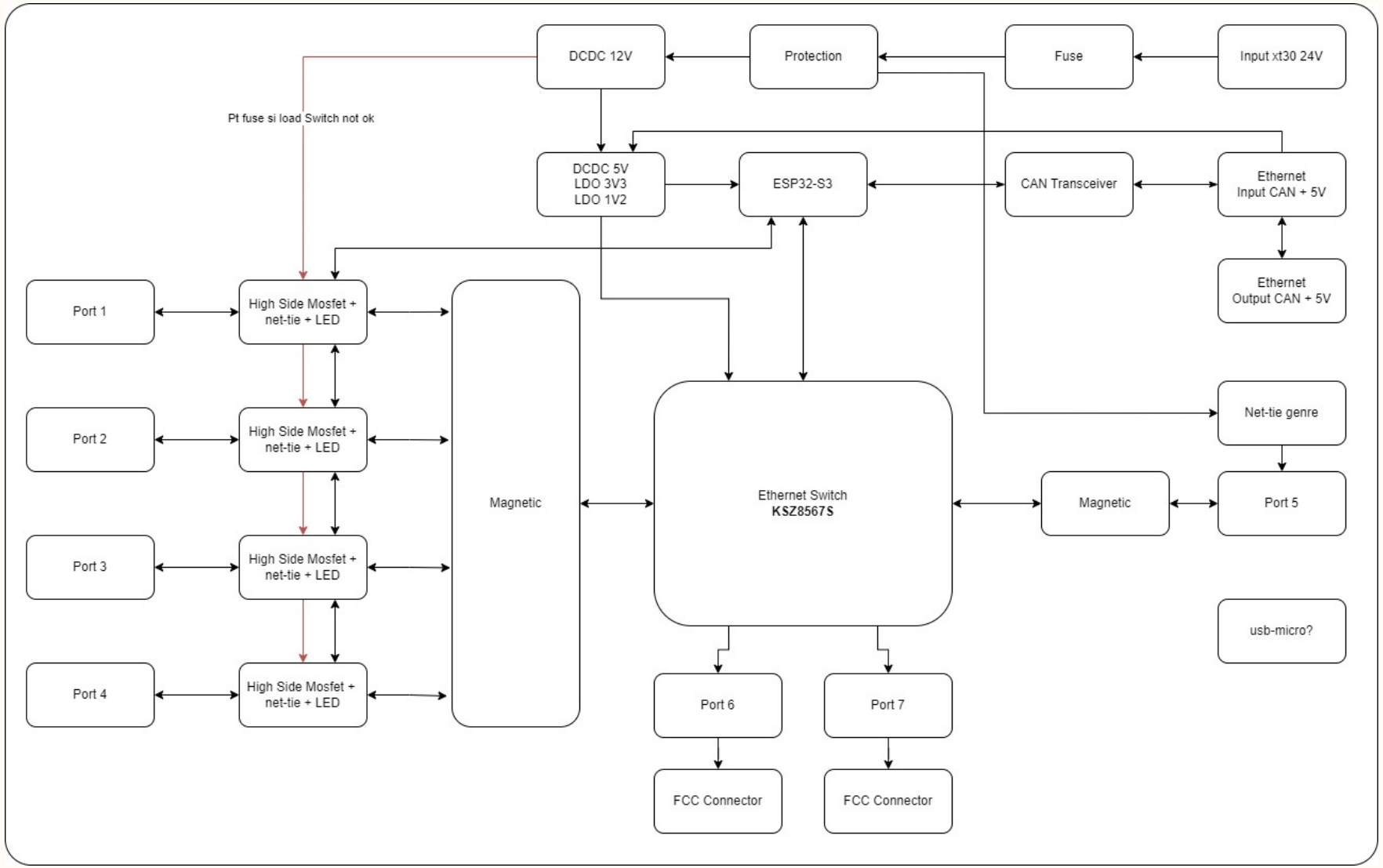


Title: roverSwitch.PrjPcb			
Size: B	Drawn by: Emile Renaud		
Date: 2026-10-07	Checked by: *		
Sheet 1 of 8	Revision: V1.0	File: Overview.SchDoc	

Power Input

P8
P 24V_RAW
N
XT30
L7
3404.2419J1
GND
Validate footprint XT30

P6
Power
24V_RAW
4pos-TH-2.54mm
GND

Reverse & OverVoltage Protection

Power
Power
Power
24V
Q1
AONR21307
Q2
AONR21307
DZ1
6.2V
R1
10k
Q3
SSM3J332R
DZ3
7V
R6
10k
R7
10k
GND

DCDC 3V3

P10 3pos-TH-2.54mm
12V
5V_USB
U1
VIN
BOOT
PHASE
FB
EN
GND
RT8258GJ6
C1
10nF
25V
C2
10uF
35V
L1
10uH
1.5A
DZ2
30V
PWR
PWR
3V3
R49
100k
R50
32.4k
C3
22uF
16V
R4
1k
LED1
GND

LDO 2V5

160mA
3V3
U3
IN
OUT
EN
GND
NC
TLV75725PDBVR
C74
10V
10uF
R51
10k
C75
10V
10uF
2V5
J20
GND

LDO 1V2

500mA
3V3
U4
IN
OUT
OUT
GND
TLV1117LV12DCYR
C76
25V
1uF
C77
25V
1uF
1V2
J21
GND

Fiducials

F1
F2
F3

DCDC 12V

3A
24V
U5
VIN
Boot
EN
SW
NC
NC
GND
GND
FB
RT6285GSP
R52
10k
C78
50V
10nF
104CDMCCDS-220MC
L8
22uH
C79
50V
100uF
C81
25V
22uF
D5
CDBA540-HF
C82
50V
82pF
R53
100k
R54
8.87k
C83
25V
22uF
C84
35V
220uF
R75
1k
LED17
12V
J22
GND

Figure 1. Output Voltage Setting

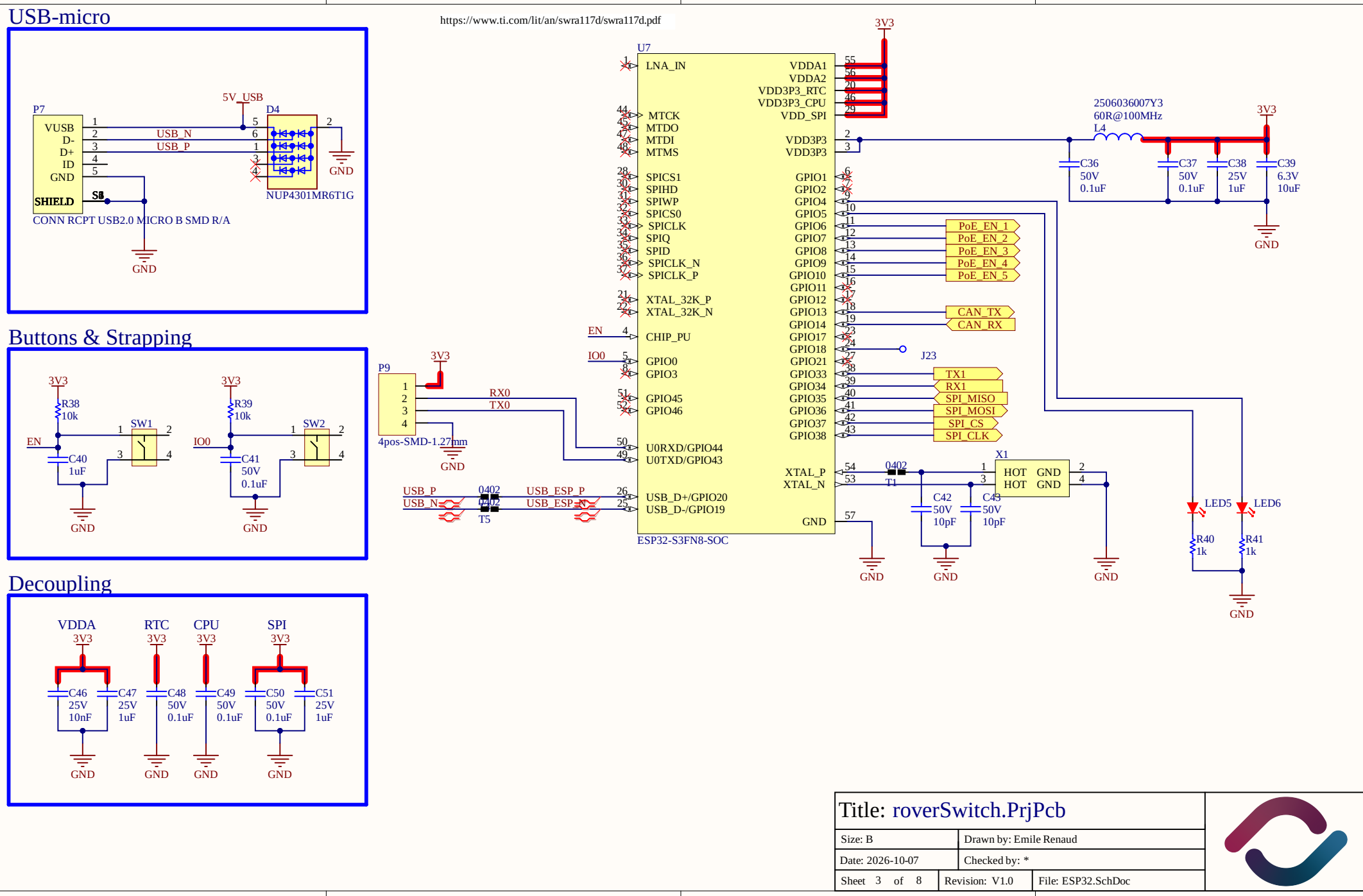
The output voltage is set by an external resistive divider according to the following equation :

$V_{OUT} = V_{REF}\left(1+\frac{R1}{R2}\right)$

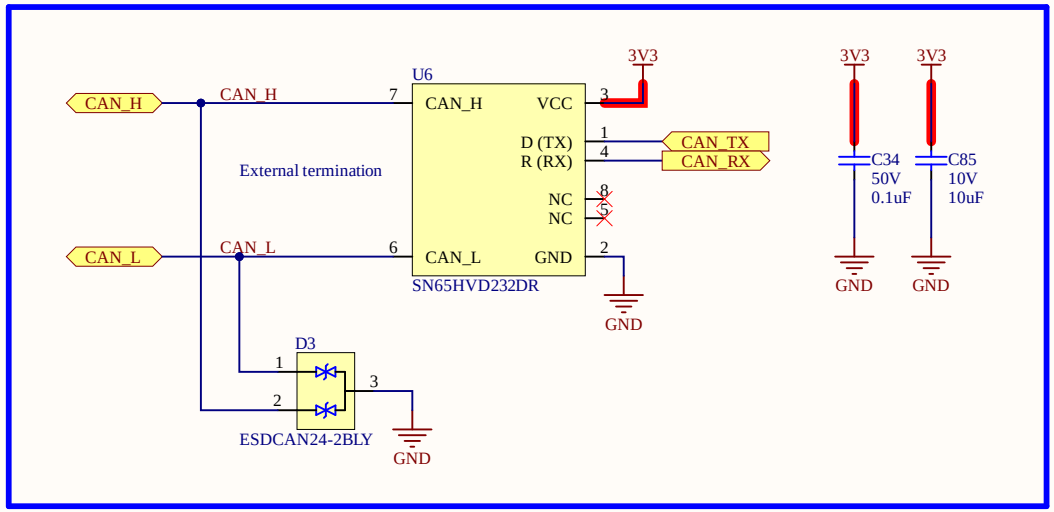
Where V_{REF} is the reference voltage (1.222V typ.).

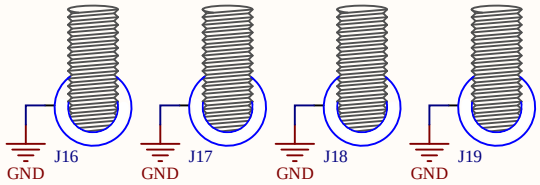
Where R1 = 100kΩ.

Title: roverSwitch.PrjPcb			
Size: B	Drawn by: Emile Renaud		
Date: 2026-10-27	Checked by: *		
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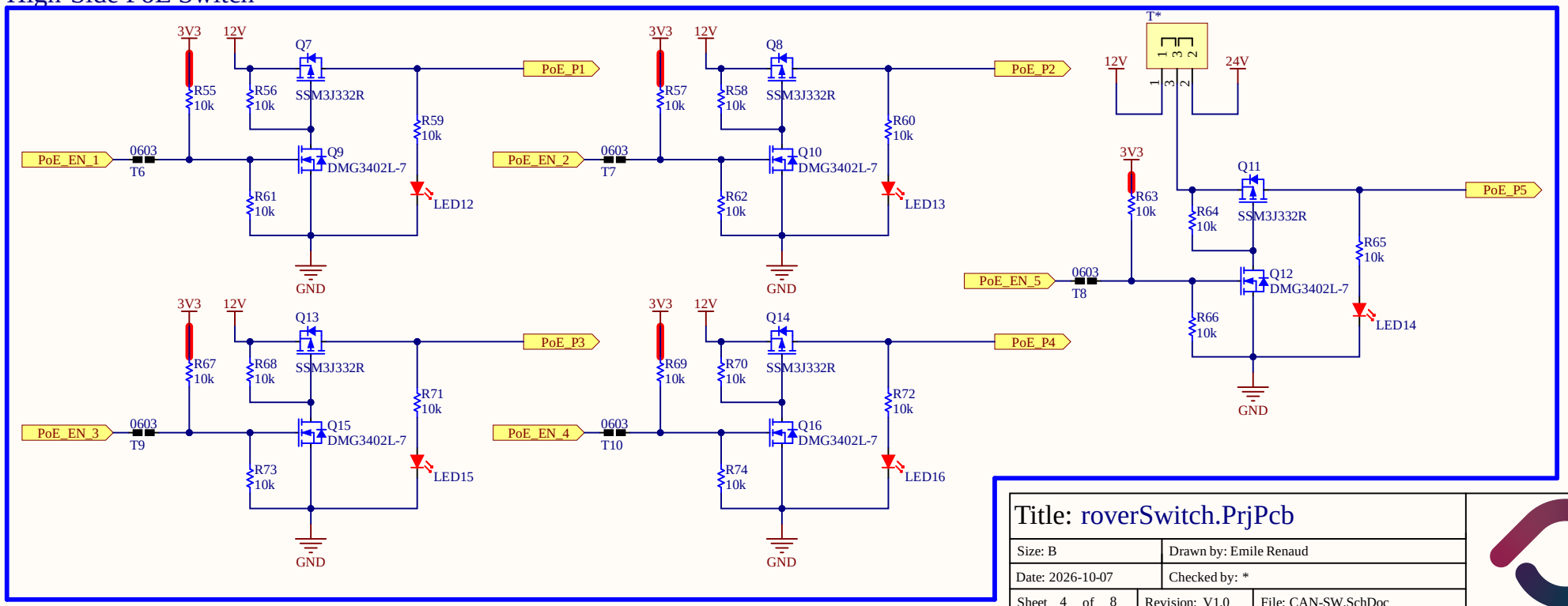

USB-micro
https://www.ti.com/lit/an/swra117d/swra117d.pdf
P7
VUSB
D-
D+
ID
GND
SHIELD
USB_N
USB_P
5V_USB
D4
NUP4301MR6T1G
CONN RCPT USB2.0 MICRO B SMD R/A
Buttons & Strapping
R38 10k
R39 10k
SW1
SW2
EN
IO0
C40 1uF
C41 50V 0.1uF
Decoupling
VDDA
RTC
CPU
SPI
C46 25V 10nF
C47 25V 1uF
C48 50V 0.1uF
C49 50V 0.1uF
C50 50V 0.1uF
C51 25V 1uF
U7
ESP32-S3FN8-SOC
LNA_IN
MTCK
MTDO
MTDI
MTMS
SPICS1
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
SPICLK_N
SPICLK_P
XTAL_32K_P
XTAL_32K_N
CHIP_PU
GPIO0
GPIO3
GPIO45
GPIO46
U0RXD/GPIO44
U0TXD/GPIO43
USB_D+/GPIO20
USB_D-/GPIO19
VDDA1
VDDA2
VDD3P3_RTC
VDD3P3_CPU
VDD_SPI
VDD3P3
GPIO1
GPIO2
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO21
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
XTAL_P
XTAL_N
PoE_EN_1
PoE_EN_2
PoE_EN_3
PoE_EN_4
PoE_EN_5
CAN_TX
CAN_RX
J23
TX1
RX1
SPI_MISO
SPI_MOSI
SPI_CS
SPI_CLK
P9
4pos-SMD-1.27mm
RX0
TX0
T5
T1
X1
HOT GND
HOT GND
2506036007Y3
60R@100MHz
L4
C36 50V 0.1uF
C37 50V 0.1uF
C38 25V 1uF
C39 6.3V 10uF
C42 50V 10pF
C43 50V 10pF
LED5
LED6
R40 1k
R41 1k
Title: roverSwitch.PrjPcb
Size: B
Drawn by: Emile Renaud
Date: 2026-10-07
Checked by: *
Sheet 3 of 8
Revision: V1.0
File: ESP32.SchDoc

CAN



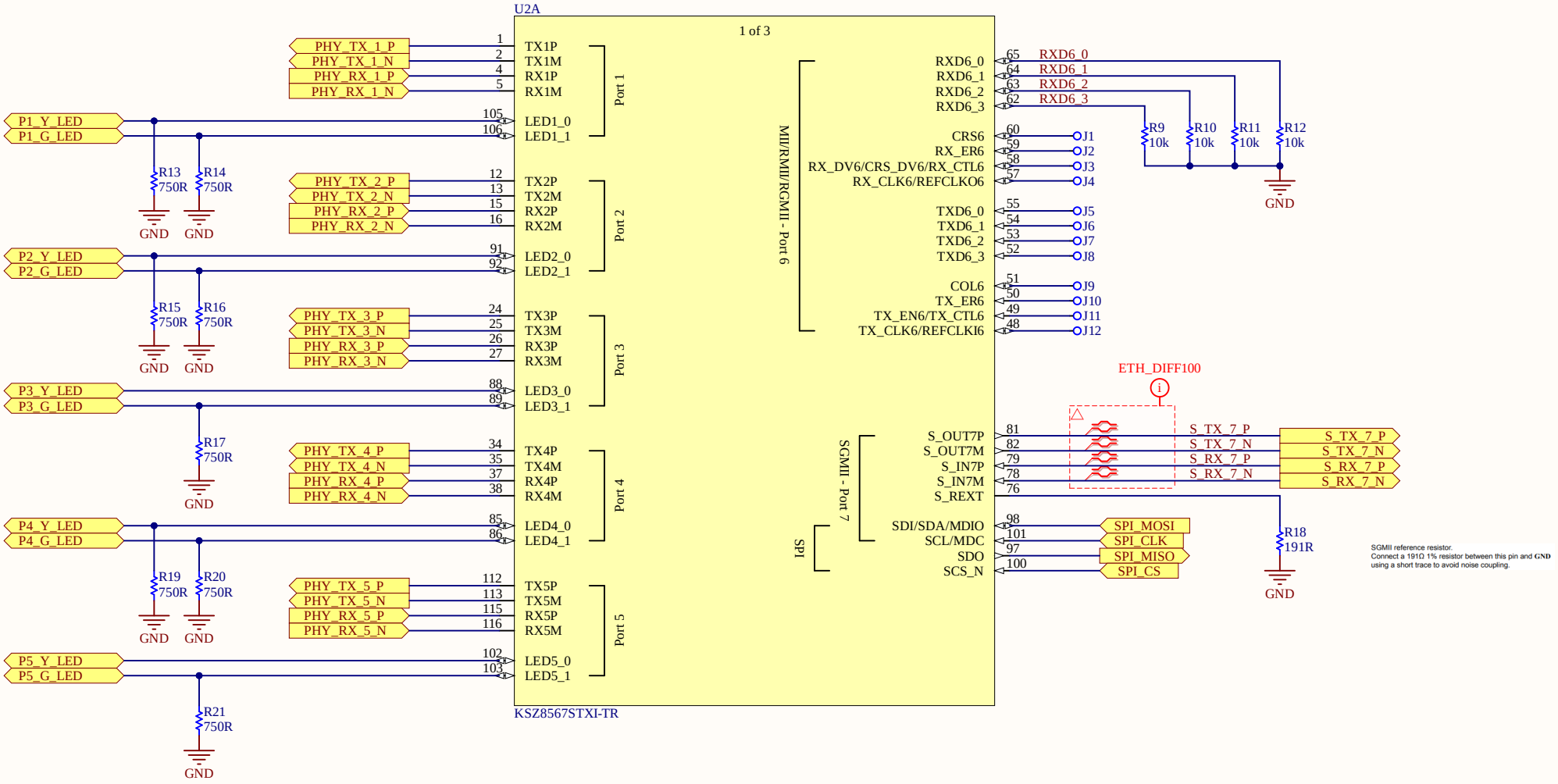


High-Side PoE Switch



Title: roverSwitch.PrjPcb			
Size: B	Drawn by: Emile Renaud		
Date: 2026-10-07	Checked by: *		
Sheet 4 of 8	Revision: V1.0	File: CAN-SW.SchDoc	

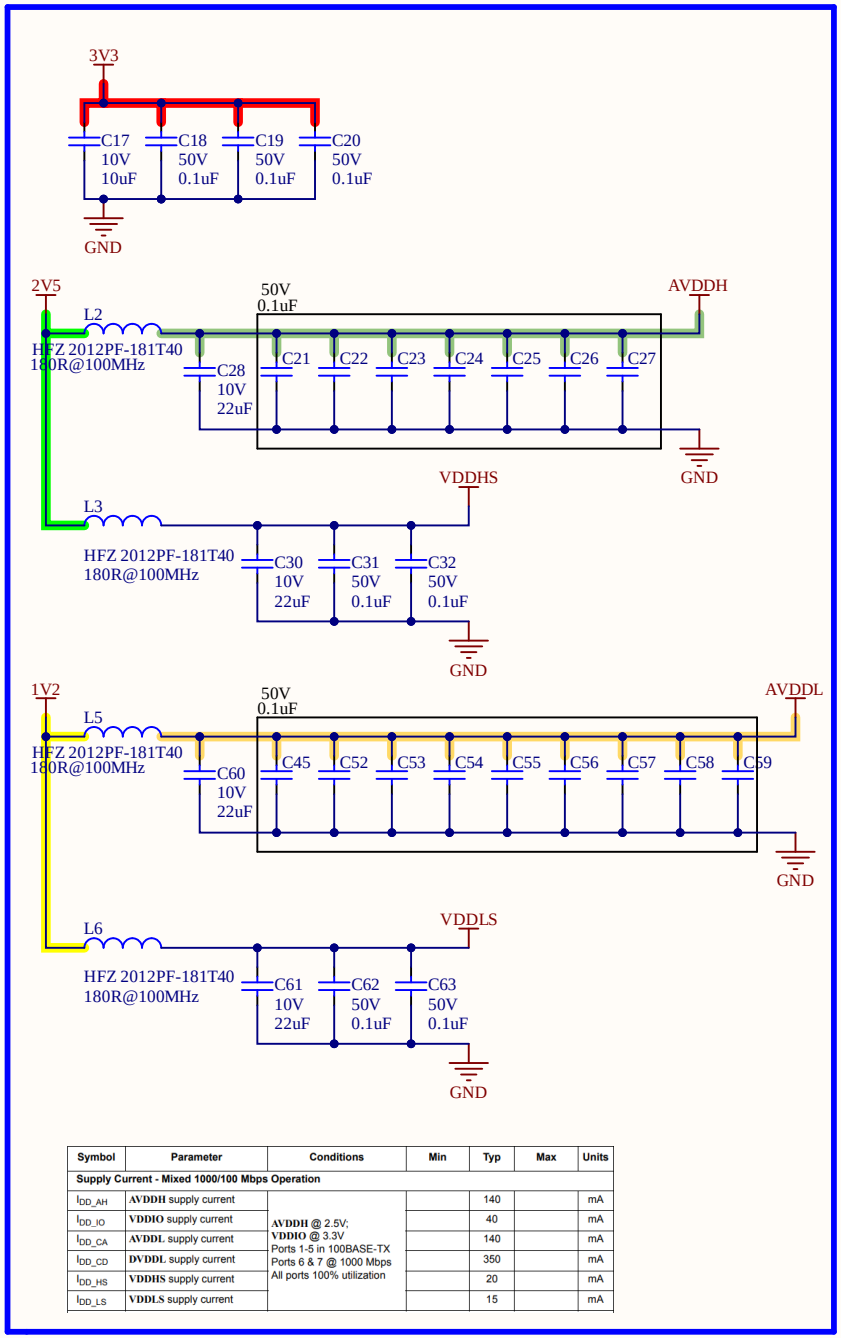
Ethernet IC - Ports



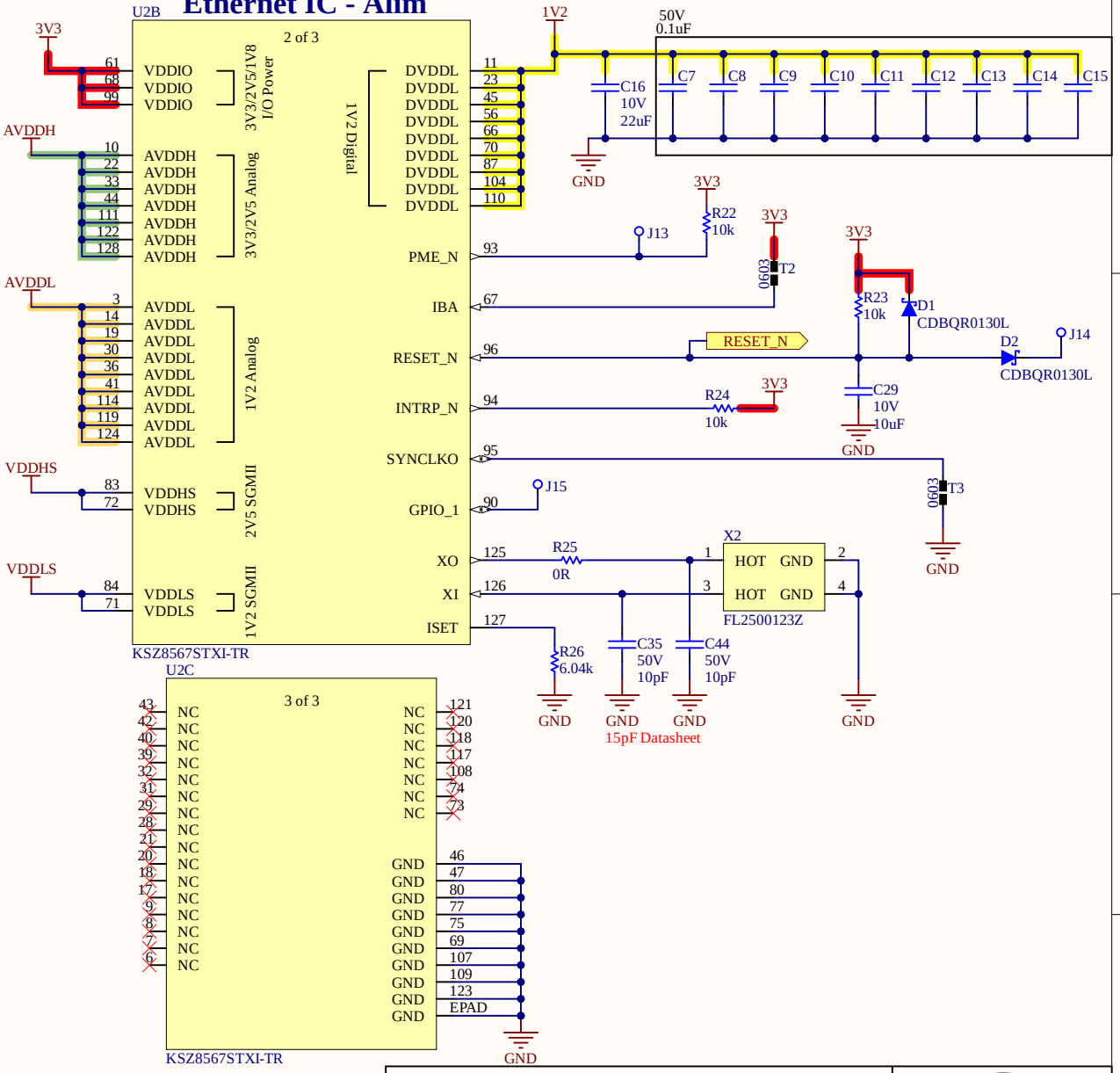
Title: roverSwitch.PrjPcb			
Size: B	Drawn by: Emile Renaud		
Date: 2026-10-07	Checked by: *		
Sheet 5 of 8	Revision: V1.0	File: KSZ_Switch.SchDoc	

Ethernet IC - Alim

Decoupling Capacitors

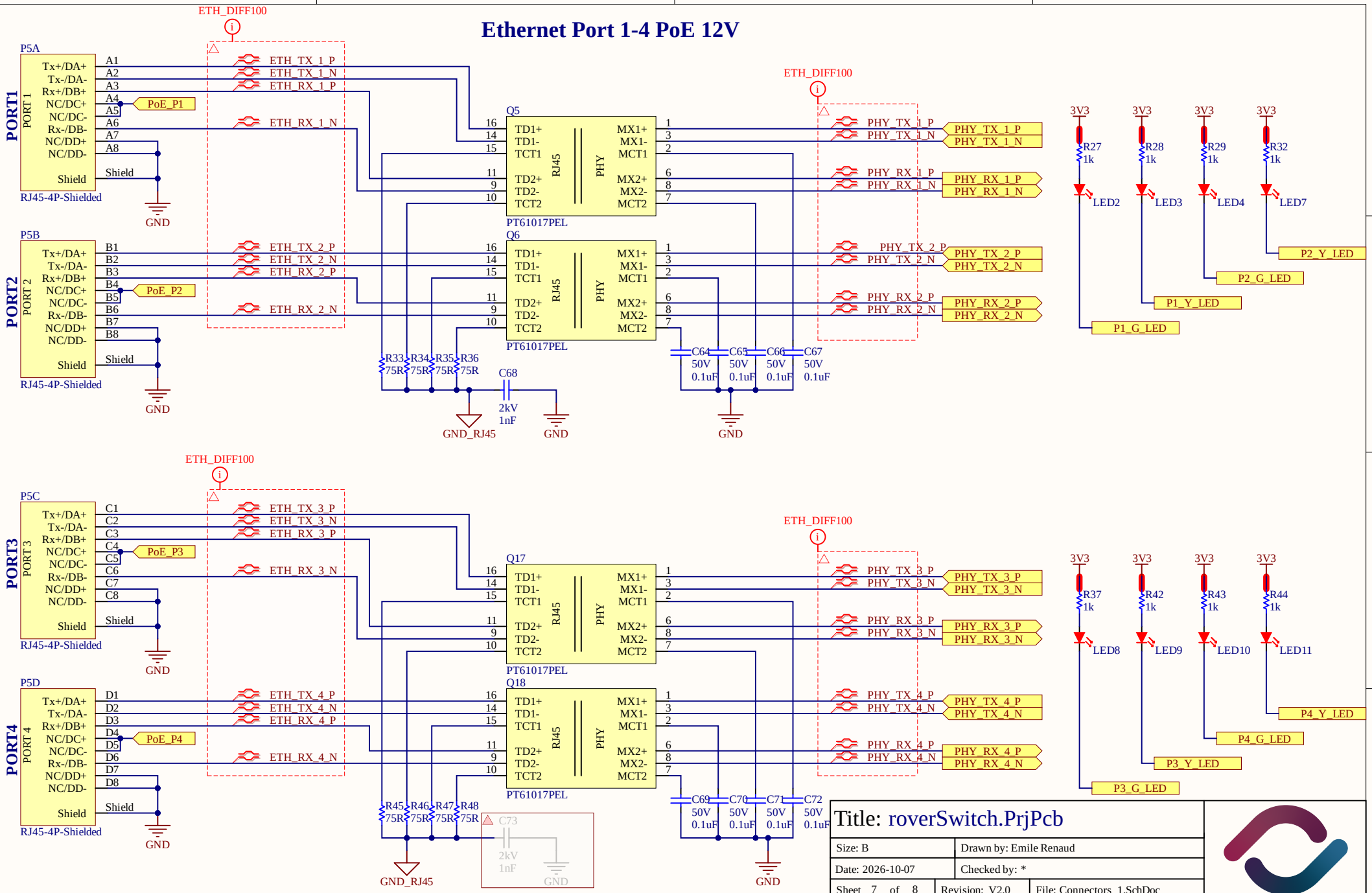


Symbol	Parameter	Conditions	Min	Typ	Max	Units
Supply Current - Mixed 1000/100 Mbps Operation						
I_{DD_AH}	**AVDDH** supply current	**AVDDH** @ 2.5V; **VDDIO** @ 3.3V Ports 1-5 in 100BASE-TX Ports 6 & 7 @ 1000 Mbps All ports 100% utilization		140		mA
I_{DD_IO}	**VDDIO** supply current			40		mA
I_{DD_CA}	**AVDDL** supply current			140		mA
I_{DD_CD}	**DVDDL** supply current			350		mA
I_{DD_HS}	**VDDHS** supply current			20		mA
I_{DD_LS}	**VDDLS** supply current			15		mA

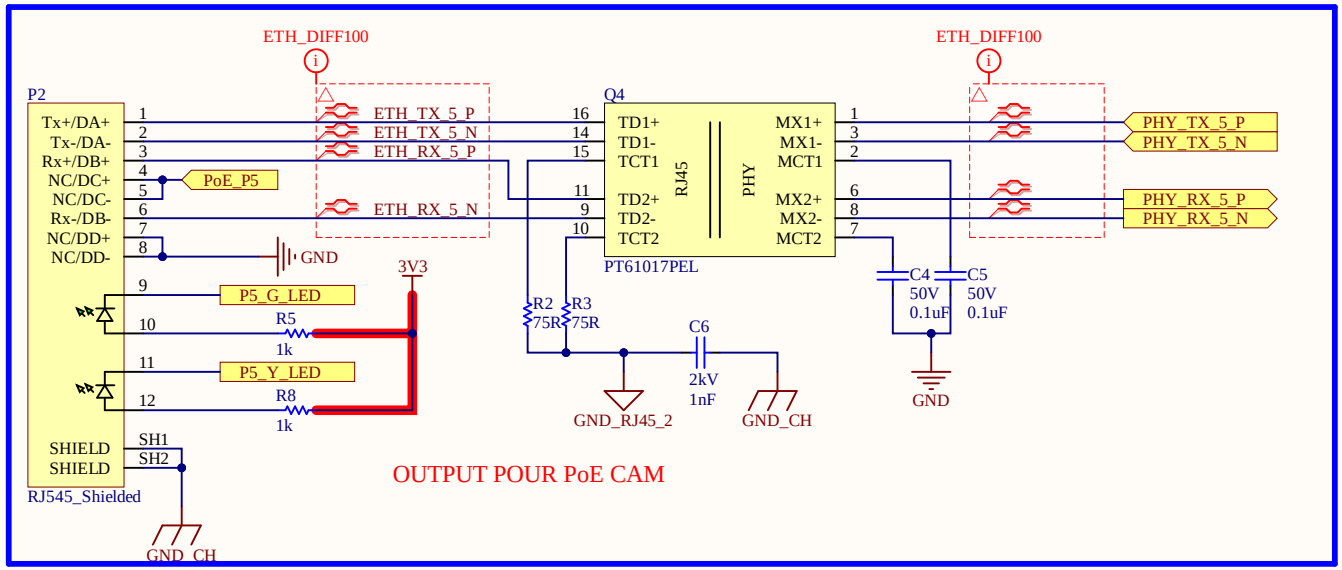


Title: roverSwitch.PrjPcb	
Size: B	Drawn by: Emile Renaud
Date: 2026-10-08	Checked by: *
Sheet 6 of 8	Revision: V1.0 — File: KSZ_Switch_Control.SchDoc

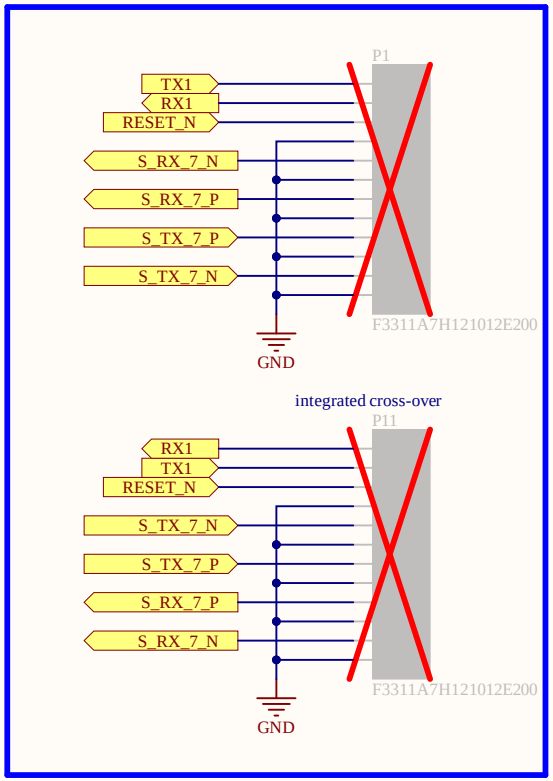
Ethernet Port 1-4 PoE 12V



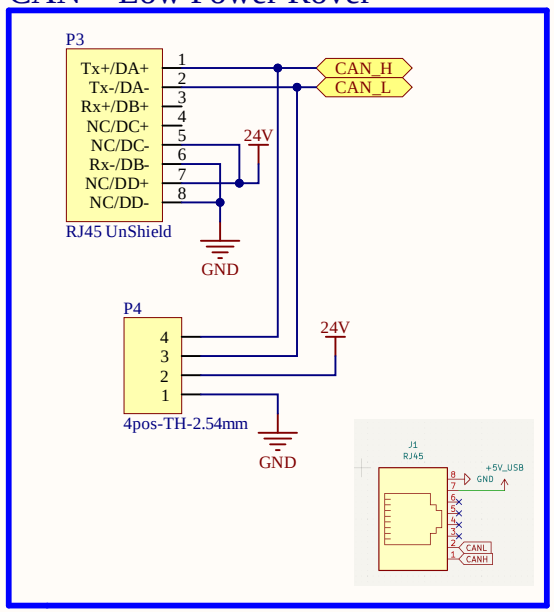
Port 5 - PoE 12/24



FFC Connector Ethernet



CAN + Low Power Rover



Title: roverSwitch.PrjPcb			
Size: B	Drawn by: Emile Renaud		
Date: 2026-10-14	Checked by: *		
Sheet 8 of 8	Revision: V1.0	File: Connectors_2.SchDoc	

Board Stack Report